

Low-rank Approximations in Transcorrelated Integrals

Ke Liao

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1 Introduction

The transcorrelated method combined with density-fitting approximation provides an efficient approach to electronic structure calculations.

2 Theoretical Framework

2.1 Transcorrelation Theory

The transcorrelated Hamiltonian is a similarity transformation on the original Hamiltonian in real space:

$$\bar{H} = e^{-\hat{\tau}} H e^{\hat{\tau}} \quad (1)$$

where $e^{\hat{\tau}}$ is the so-called Jastrow factor and $\hat{\tau}$ is the correlation factor operator. A simple form of $\hat{\tau}$ is given by the two-body Jastrow factor:

$$\hat{\tau} = \frac{1}{2} \sum_{ij}^N u(r_{ij}) \quad (2)$$

where $f(r_{ij})$ is a correlation function that depends on the inter-electron distance r_{ij} .

The TC Hamiltonian can be expanded using the Baker-Campbell-Hausdorff (BCH) formula and it terminates exactly at the second commutator, gener-

ating additional two-body and three-body terms:

$$\begin{aligned}
\bar{H} &= \hat{H} + [\hat{\tau}, \hat{H}] + \frac{1}{2}[[\hat{\tau}, \hat{H}], \hat{\tau}] \\
&= \hat{H} - \frac{1}{2} \sum_{ij} \mathcal{P}(\mathbf{r}_i, \mathbf{r}_j) \left[\frac{1}{2}(\nabla_i^2 u(\mathbf{r}_i, \mathbf{r}_j) + (\nabla_i u(\mathbf{r}_i, \mathbf{r}_j))^2) + \nabla_i u(\mathbf{r}_i, \mathbf{r}_j) \cdot \nabla_i \right] \\
&\quad - \frac{1}{6} \sum_{ijk} \mathcal{P}(\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k) \nabla_i u(\mathbf{r}_i, \mathbf{r}_j) \cdot \nabla_i u(\mathbf{r}_i, \mathbf{r}_k)
\end{aligned} \tag{3}$$

Using a set of orbitals $\{\phi_p\}$, the TC Hamiltonian can be expressed in the second quantization form:

$$\bar{H} = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} (V_{rs}^{pq} - K_{rs}^{pq}) a_p^\dagger a_q^\dagger a_s a_r - \frac{1}{6} L_{stu}^{pqr} a_p^\dagger a_q^\dagger a_r^\dagger a_u a_t a_s \tag{4}$$

where we use the Einstein summation convention. We have the K and L tensors defined as:

$$\begin{aligned}
K_{rs}^{pq} &= \hat{\mathcal{P}}_{12} \left[\langle \phi_p(\mathbf{r}_1) \phi_q(\mathbf{r}_2) | \frac{1}{2}(\nabla_1^2 u(\mathbf{r}_1, \mathbf{r}_2) + (\nabla_1 u(\mathbf{r}_1, \mathbf{r}_2))^2) + \nabla_1 u(\mathbf{r}_1, \mathbf{r}_2) \cdot \nabla_1 | \phi_r(\mathbf{r}_1) \phi_s(\mathbf{r}_2) \rangle \right] \\
L_{stu}^{pqr} &= \hat{\mathcal{P}}_{123} \langle \phi_p(\mathbf{r}_1) \phi_q(\mathbf{r}_2) \phi_r(\mathbf{r}_3) | \nabla_1 u(\mathbf{r}_1, \mathbf{r}_2) \cdot \nabla_1 u(\mathbf{r}_1, \mathbf{r}_3) | \phi_s(\mathbf{r}_1) \phi_t(\mathbf{r}_2) \phi_u(\mathbf{r}_3) \rangle
\end{aligned} \tag{5}$$

2.2 Interpolative Density-Fitting Approximation

We see that the main quantity to approximate in the K and L tensors is the three-center integrals:

$$\begin{aligned}
\mathbf{G}(\mathbf{r}_1, \mathbf{r}_2) &= \nabla_1 u(\mathbf{r}_1, \mathbf{r}_2) \\
\rho_q^p(\mathbf{r}) &= \phi_p^*(\mathbf{r}) \phi_q(\mathbf{r}) \\
\mathbf{W}_q^p(\mathbf{r}) &= \phi_q(\mathbf{r}) \nabla_1 \phi_p^*(\mathbf{r})
\end{aligned} \tag{6}$$

Let's first examine if the $\mathbf{G}(\mathbf{r}_1, \mathbf{r}_2)$ can be approximated by a low-rank tensor.

3 Implementation of DF in TC Integrals

3.1 Three-Center Integrals

The three-center integrals in the context of TC theory:

$$(pq|\alpha) = \int \phi_p(\mathbf{r}_1)\phi_q(\mathbf{r}_1)P_\alpha(\mathbf{r}_2)\frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|}d\mathbf{r}_1d\mathbf{r}_2 \quad (7)$$

3.2 Modified TC Integrals

Discussion of how density-fitting modifies the TC integrals...

4 Computational Efficiency

Analysis of computational scaling and memory requirements...

5 Conclusions

Summary of the advantages and potential applications...